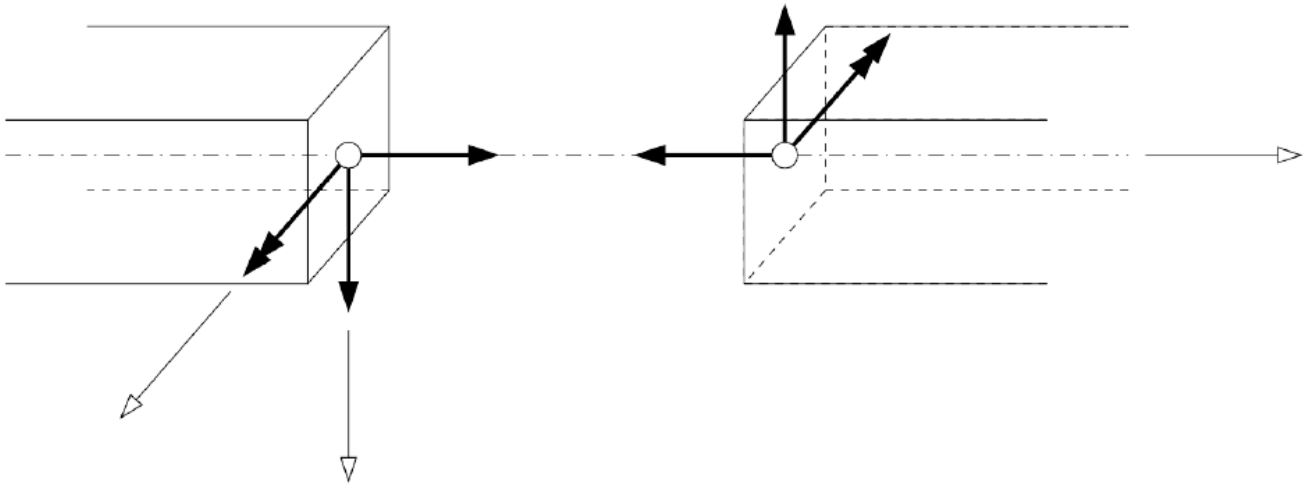
Engineering Mechanics I

Lecture



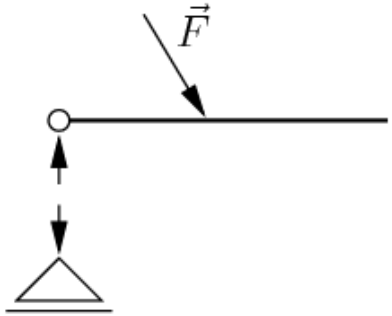
7.3. Stress resultants of a beam

- So far we have only considered **external forces** on the beam, e.g. loads and bearing reactions. These must satisfy the equilibrium of the overall system (total beam).

- However, the equilibrium must also be satisfied on **subsystems**. Subsystems (sections) from the beam can be obtained by cutting the beam.

- The internal forces required at the interfaces to keep sections of the beam in equilibrium are called **stress resultants**.

7.3. Stress resultants of a beam

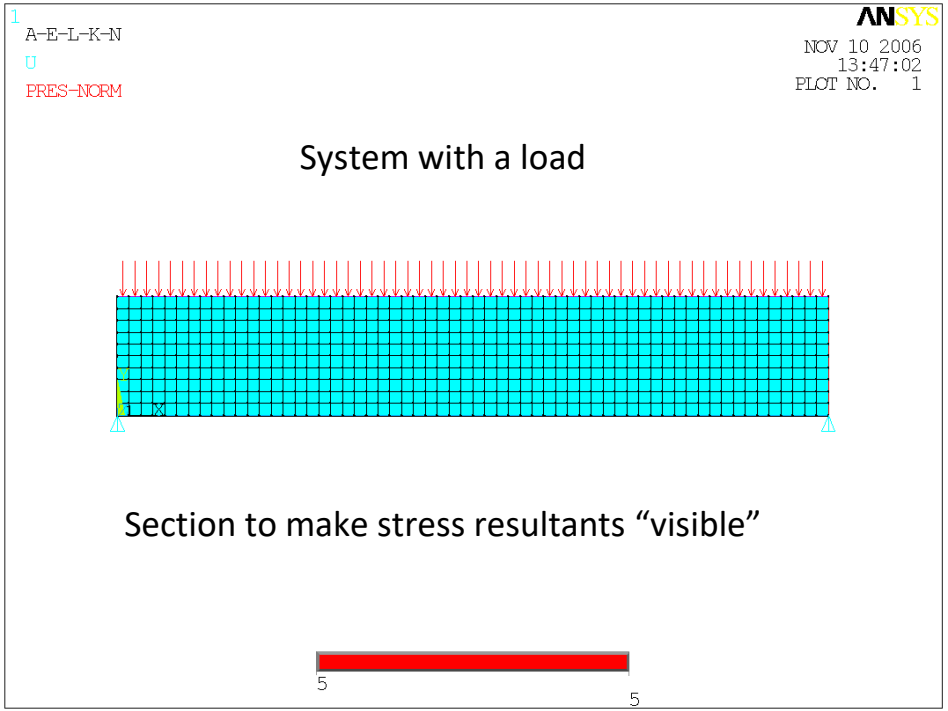


Box 7.10

- **Normal force** $\vec{N}$, acts in the direction of the bar axis
- **Shear force** $\vec{Q}$, runs perpendicular to the bar axis
- **(Bending) Moment** $\vec{M}$, prevents rotations of the beam part

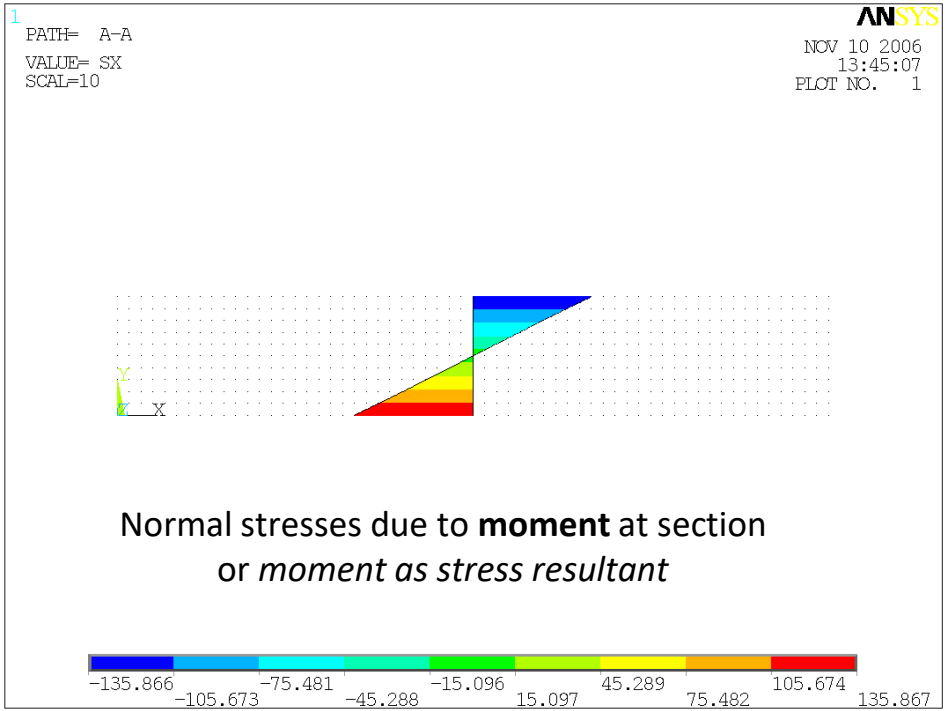
7.3. Stress resultants of a beam

Finite element calculation - beam with line load



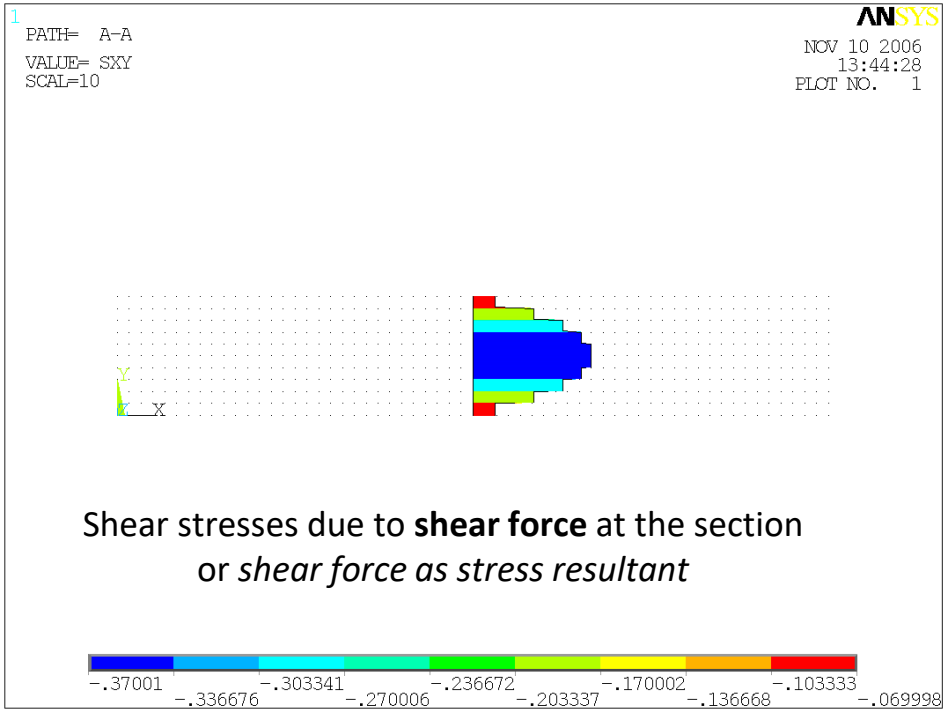
7.3. Stress resultants of a beam

Finite element calculation - beam with line load



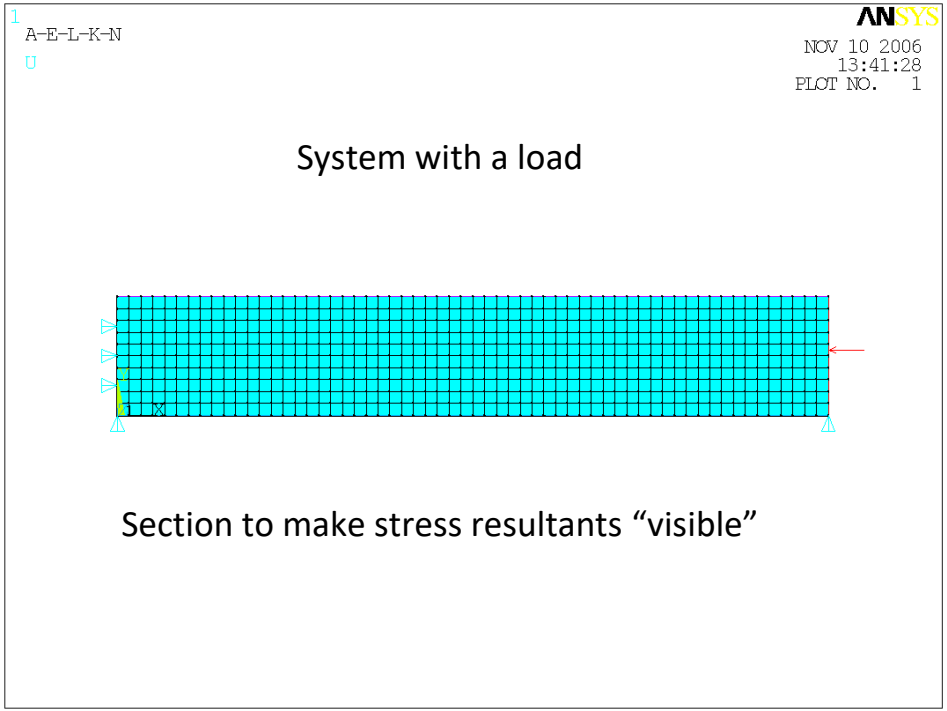
7.3. Stress resultants of a beam

Finite element calculation - beam with line load



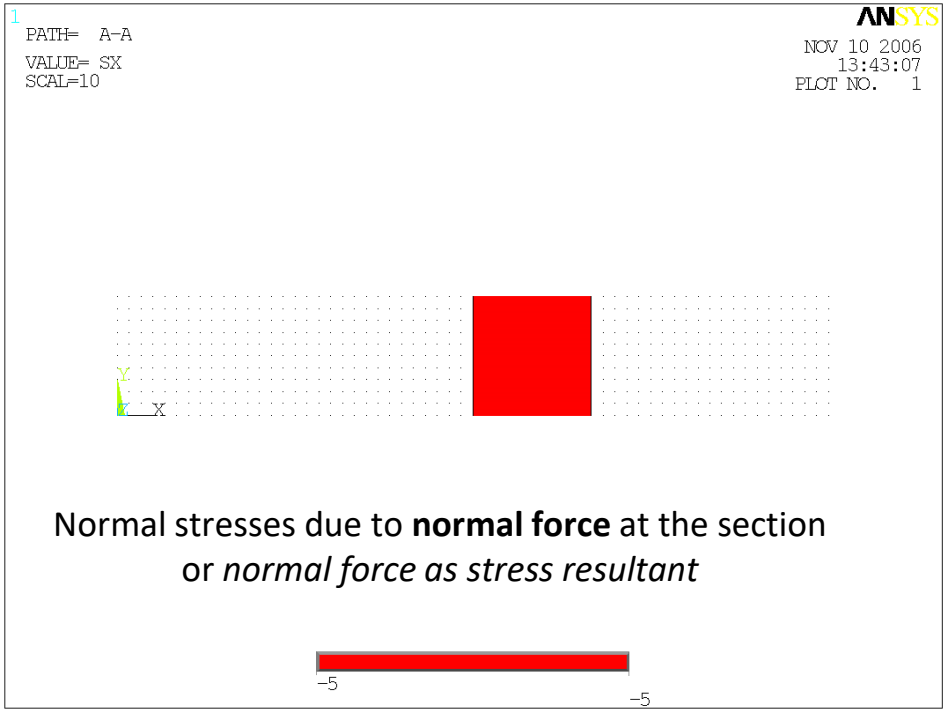
7.3. Stress resultants of a beam

Finite element calculation - beam with line load



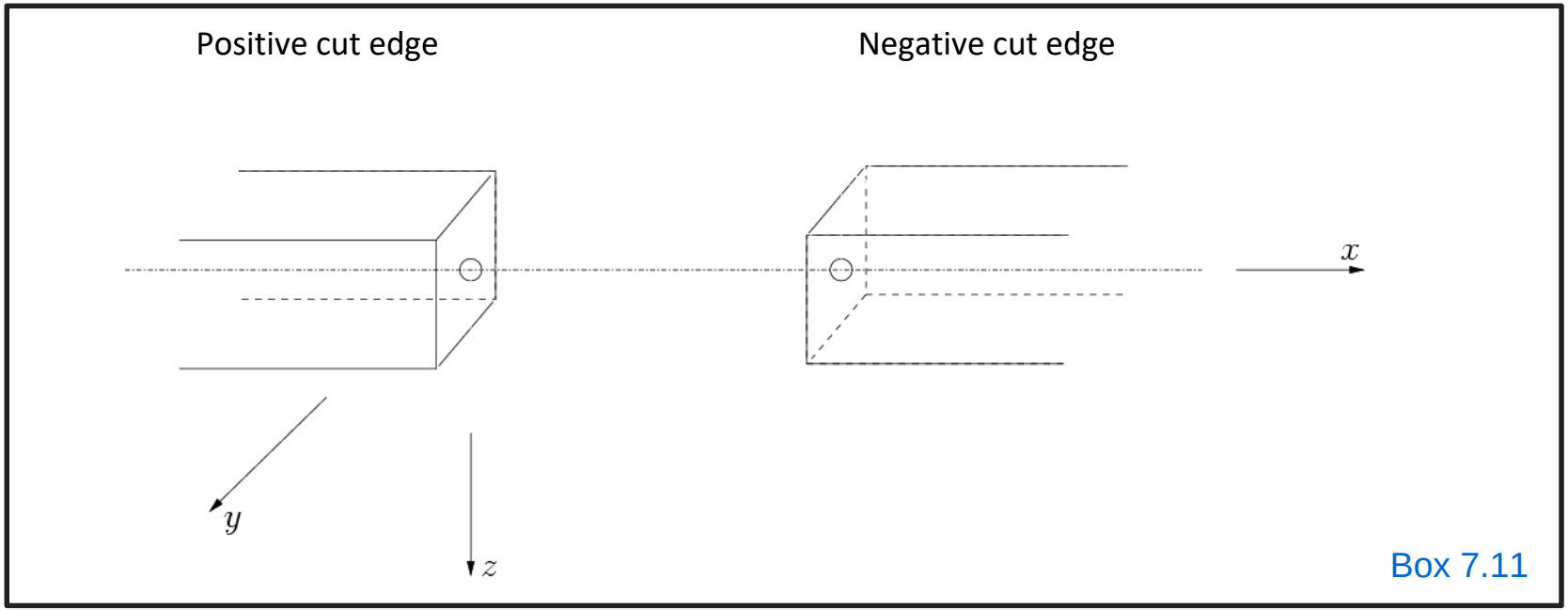
7.3. Stress resultants of a beam

Finite element calculation - beam with line load



7.3. Stress resultants of a beam

Sign definitions for stress resultants



Box 7.11

7.3. Stress resultants of a beam

Cut edge definitions:

- Positive cut edge: positive x-axis runs out
- Negative cut edge: positive x-axis runs in

Normal forces:

- Positive normal forces act in the positive x-direction on the positive cut edge, and in the negative x-direction on the negative cut enge.

  positive normal force <---> Tractive force

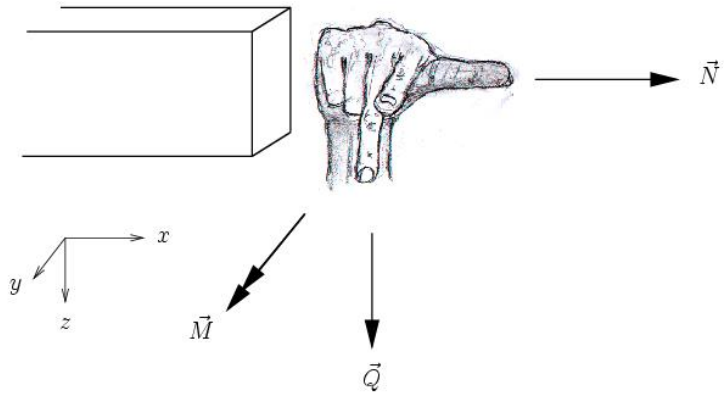
  negative normal force <-> Compression force

Shear forces:

- Positive shear forces point in the positive z-direction at the positive cut edge, and correspondingly in the negative z-direction at the negative edge.

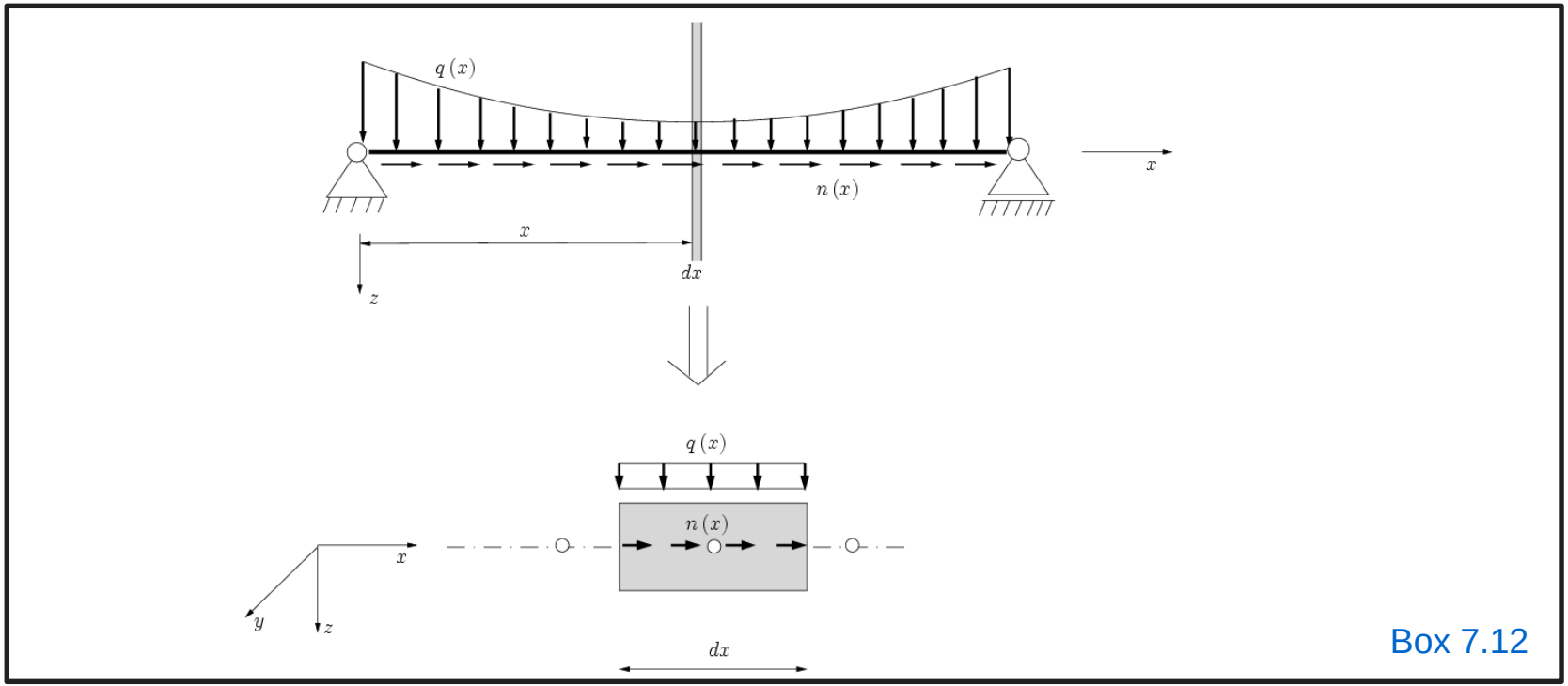
7.3. Stress resultants of a beam

- Bending moment is defined at the positive cutting edge so that it forms a right system



- Positive bending moments generate tensile stresses on the positive z-side

7.4. Differential relations on the straight beam



Box 7.12

7.4. Differential relations on the straight beam

Equilibrium at the infinitesimal element :

- Sum of forces in x-direction

Box 7.13

$$dN\,(x) = -n(x) \cdot dx$$

7.4. Differential relations on the straight beam

Equilibrium at the infinitesimal element :

- Sum of forces in z-direction



Box 7.14

$$dQ\,(x) = -q(x) \cdot dx$$

7.4. Differential relations on the straight beam

- Moment sum around center of diff. element

- Continuous load is considered constant and therefore does not yield a moment component

$$\sum M = 0:$$

$$M(x) + dM(x) - [Q(x) + dQ(x)]\frac{dx}{2} - Q(x)\frac{dx}{2} - M(x) = 0$$

Box 7.15

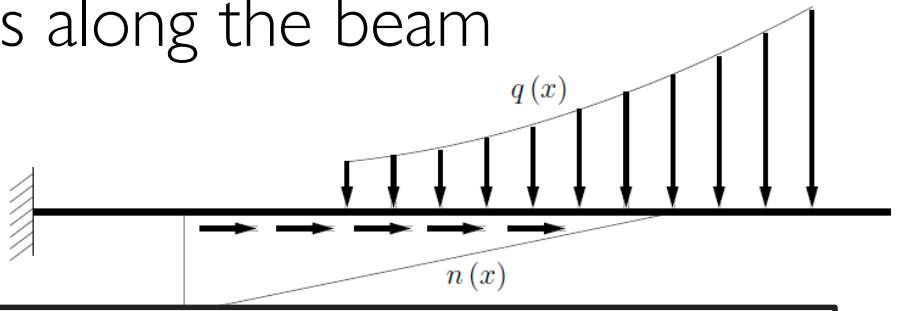
7.4. Differential relations on the straight beam

Summary

Box 7.16

7.5. Course of stress resultants along the beam

- **1. Step: Continuous load without concentrated (point) loads**



$$N(x) = \int_{x_0}^{x} \frac{dN}{dx} dx + N(x_0) =$$

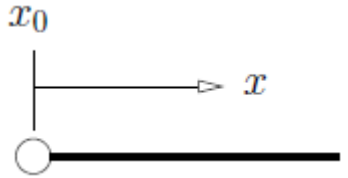
$$Q(x) = \int_{x_0}^{x} \frac{dQ}{dx} dx + Q(x_0) =$$

$$M(x) = \int_{x_0}^{x} \frac{dM}{dx} dx + M(x_0) =$$

Box 7.17

7.5. Course of stress resultants along the beam - Boundary conditions

For supports without external loads



Symbol	Support	Known without computation	Support forces (known from global equilibrium)
	Pin		
	Parallel motion		
	Fixed end		
	Free end		

Box 7.18